

Audrey Dominique

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EDUCATION

Singapore University of Technology and Design (SUTD)

Sep 2024 to May 2028 (expected)

- Bachelor of Engineering (Product and Development) — Electrical Engineering Track

AWARDS

- Baby Shark Fund Award**, Received S\$6,000 in funding support for SAND-E, a remote-controlled beach-cleaning robot transitioning toward autonomous navigation
- Top course project**, Engineering for Assistive Good and Empowerment by Design (Hane board)

ACADEMIC PROJECTS

Singapore University of Technology and Design

Design Thinking Innovation – Trash can?

Jan 2025 to May 2025

- Designed and 3D-printed mechanical components for an auto-sorting trashcan prototype.
- Assembled and connected microcontroller-based circuits with sensors and actuators to enable computer-vision-based waste sorting

3D Design Project EngageD – Hane Board

Jan 2026 to May 2026

- Collaborated with Flour Power to develop an assistive task board for neurodivergent F&B workers.
- Integrated a MOSFET and solenoid control circuits that allow the assistive task board to push puzzle pieces and provide physical feedback to users.

WORK EXPERIENCE

Logic Coder - Part-Time Coding Teacher

Sept 2025 to Nov 2025

- Taught students the fundamentals of Python programming and guided students in applying Python to simple game development Pygame

Fabrication LAB Assistant in SUTD

Aug 2025 to Nov 2025

- Assisted in 3D printing and laser cutting during student orientation training and visiting guests from China

CO-CURRICULAR ACTIVITIES

SUTD Organization for Autonomous Robotics (SOAR)

Beach Cleaning robot (SAND-E) team member

Sep 2024 to June 2026

- Designed and refined mechanical structure of trash picking mechanism, applying engineering calculations to support prototyping, structural design, and design validation.
- Implemented ROS2 in microcontrollers (micro-ROS) to integrate motors and sensors, enabling laptop-based control, SLAM mapping, RViz visualization, and basic robot navigation.
- Assembled electronics hardware modules and calculated component requirements to select suitable motors, sensors, and power systems

SOAR Challenge Organizing Committee

July 2025 to Oct 2025

- Designed and fabricated a plywood robot playfield using CAD and laser cutting
- Assembled robotics kits by soldering electronic components and wiring connections.

SUTD Student Government

SUTD Freshmen orientation creative media committee

July 2025 to Sept 2025

- Led the design and creation of event centerpiece from sketches to final prototype as a rocket
- Fabricated the main rocket structure from cardboard using a Zünd digital cutting system
- Designed custom 3D printed rocket components, integrating ESP32-controlled LED strips for interactive lighting effects.

Vision-Language-Action (VLA) Robot Arm – SIX Eyes

Jan 2026 to present

- Designed the schematic and PCB layout for a robotic leader arm in KiCad, integrating potentiometer sensing, RC filtering, JST connectors, and an ESP32-C6 SuperMini.

ADDITIONAL INFORMATION

- Technical Skills: Fusion 360, Python, ROS2, 3D printing, GitHub, Micro-ROS